

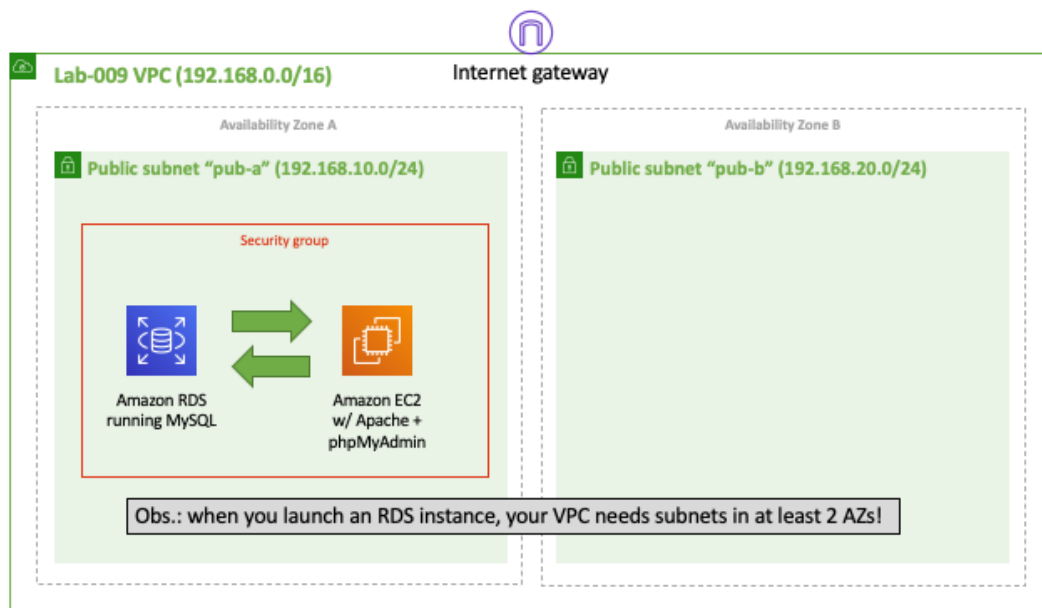
## Lab-009

### A Standalone RDS Instance Running MySQL

#### Goal

This lab illustrates how to launch a standalone RDS instance running a MySQL database. An EC2 instance will also be launched to allow access to the database using phpMyAdmin. [Here](#) is a mind map of RDS.

#### Architecture Diagram



#### Overview

The Amazon Relational Database Service (Amazon RDS) offers a managed service based on relational databases. In this lab you will be configuring RDS to use MySQL, an open source database. For simplicity, you will not configure more advanced features available in RDS, such as *multi-az deployment* or *read replicas*. To verify that RDS is working, you will configure an EC2 instance with Apache and phpMyAdmin, a free software tool written in PHP that simplifies administration of a MySQL database over the Web.

#### Step 1 - Create a Custom VPC with Two Public subnets

Start this lab by configuring an internet gateway on a custom VPC. Make sure there is a default route for your VPC's internet gateway. Make sure your public subnets are configured in distinct AZs and that auto-assign IPv4 is enabled.

#### Step 2 - Launch an RDS Instance

Below are some of the main options you should select:

- Database creation method: Standard Create
- Engine option: MySQL

- Templates: Free tier
- DB instance identifier: lab009
- Master password: 12345678 (not safe but this is just a lab)
- DB instance size: Standard classes
- Storage: default options are fine for this lab
- Multi-az deployment: if you selected the *free tier* template you should not even be able to select *multi-az deployment* options
- Connectivity: set your custom VPC to be the one where your RDS instance will run; in the additional connectivity configuration - VPC security group, choose *Create new* and name it *lab009sg*; finally for the availability zone choose the one suffixed with 1a (or the first option that appears to you)
- Additional configurations: set the initial database name to *lab009* and disable automatic backups

### Connectivity

Virtual private cloud (VPC) [Info](#)  
VPC that defines the virtual networking environment for this DB instance.  
myVPC (vpc-0f34bf2b2eba90154) ▼  
Only VPCs with a corresponding DB subnet group are listed.

ⓘ After a database is created, you can't change the VPC selection.

▼ Additional connectivity configuration

Subnet group [Info](#)  
DB subnet group that defines which subnets and IP ranges the DB instance can use in the VPC you selected.  
default-vpc-0f34bf2b2eba90154 ▼

Publicly accessible [Info](#)

☐ Yes  
Amazon EC2 instances and devices outside the VPC can connect to your database. Choose one or more VPC security groups that specify which EC2 instances and devices inside the VPC can connect to the database.

☒ No  
RDS will not assign a public IP address to the database. Only Amazon EC2 instances and devices inside the VPC can connect to your database.

VPC security group  
Choose one or more RDS security groups to allow access to your database. Ensure that the security group rules allow incoming traffic from EC2 instances and devices outside your VPC. (Security groups are required for publicly accessible databases.)

☐ Choose existing  
Choose existing VPC security groups

☒ Create new  
Create new VPC security group

New VPC security group name  
lab009sg

Availability Zone [Info](#)  
us-east-1a ▼

### Step 3 - Launch an EC2 Instance

Configure an EC2 instance in the public subnet in the same AZ where your RDS instance was launched. Use the [user-data](#). Create a security group that allows SSH and HTTP access to your instance from anywhere.

### Step 4 - Modify RDS's Security Group

Modify the inbound rule of the RDS's security group to enable access from the EC2's instance.

EC2 > Security Groups

Security Groups (1/5) [Info](#)

Filter security groups

|                                     | Name | Security group ID    | Security group name | VPC ID                | Description                | Owner        | Inbound rules count | Outbound rules count |
|-------------------------------------|------|----------------------|---------------------|-----------------------|----------------------------|--------------|---------------------|----------------------|
| <input checked="" type="checkbox"/> | -    | sg-0bd775c12d5928a3b | lab009sg            | vpc-080c2ae1ce3f75272 | Created by RDS manag...    | 939096940193 | 1 Permission entry  | 1 Permission entry   |
| <input type="checkbox"/>            | -    | sg-0e7faa8400188b0b5 | default             | vpc-080c2ae1ce3f75272 | default VPC security gr... | 939096940193 | 1 Permission entry  | 1 Permission entry   |
| <input type="checkbox"/>            | -    | sg-0ed3a18a14c6c6e4f | web-acc...          | vpc-080c2ae1ce3f75272 | web-acc...                 | 939096940193 | 1 Permission entry  | 1 Permission entry   |

sg-0bd775c12d5928a3b - lab009sg

Details **Inbound rules** Outbound rules Tags

**Inbound rules** [Edit inbound rules](#)

| Type         | Protocol | Port range | Source         | Description - optional |
|--------------|----------|------------|----------------|------------------------|
| MYSQL/Aurora | TCP      | 3306       | 76.25.79.88/32 | -                      |

EC2 > Security Groups > sg-0bd775c12d5928a3b - lab009sg > Edit inbound rules

**Edit inbound rules** [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

**Inbound rules** [Info](#)

Type [Info](#) Protocol [Info](#) Port range [Info](#) Source [Info](#) Description - optional [Info](#)

MYSQL/Aurora TCP 3306 Custom

76.25.79.88/32 X

192.168.10.248/32 X

private IP of the EC2 instance

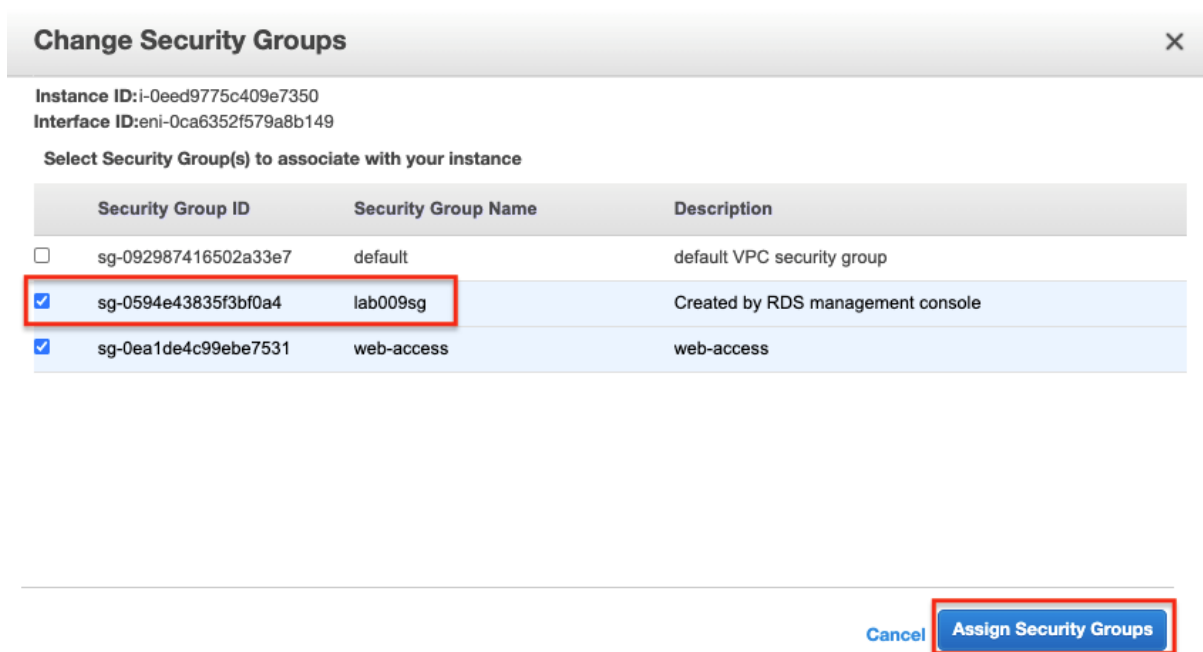
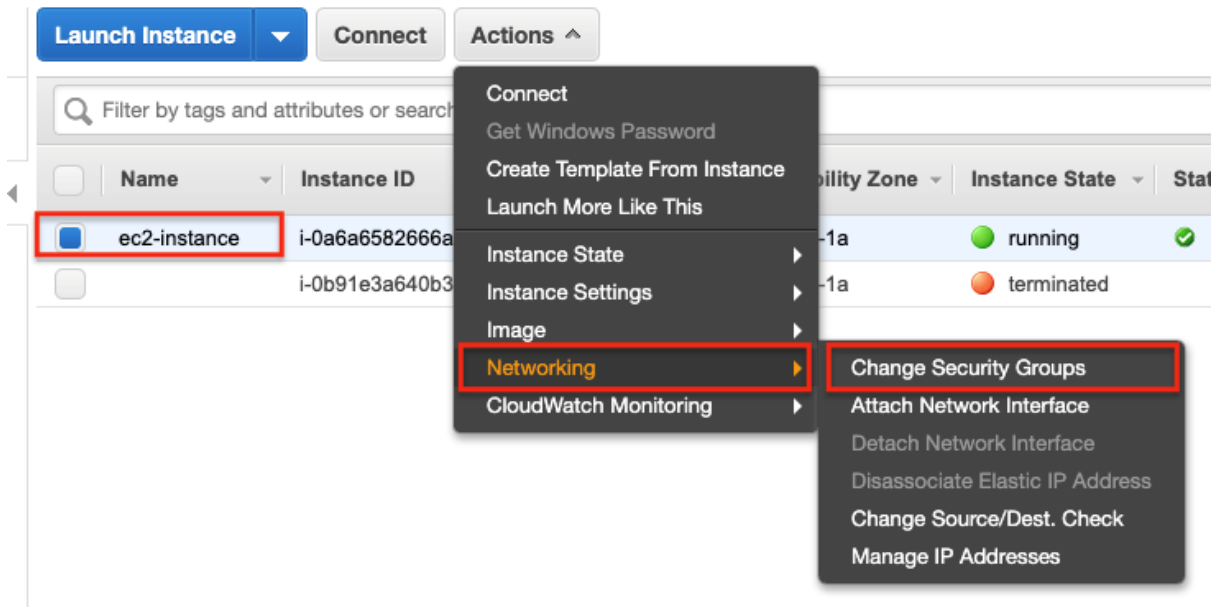
Add rule

NOTE: Any edits made on existing rules will result in the edited rule being deleted and a new rule created with the new details. This will cause traffic that depends on that rule to be dropped for a very brief period of time until the new rule can be created.

Cancel Preview changes **Save rules**

## Step 5 - Apply RDS's Security Group to EC2 Instance

In order for your EC2 instance to be able to access the RDS instance it has to be in the same security group created in step 2.



## Step 5 - Configure phpMyAdmin

- Connect to your EC2 instance via SSH
- Run the following commands (requires privileged access):

```
cd /var/www/html
tar xvf phpMyAdmin-latest-all-languages.tar.gz
rm -rf phpMyAdmin-latest-all-languages.tar.gz
ln -s phpMyAdmin-5.0.2-all-languages phpMyAdmin
cd phpMyAdmin
cp config.sample.inc.php config.inc.php
```

Edit *config.inc.php* and search for a reference to *localhost*. Replace it with your RDS instance endpoint.

## lab-009

## Summary

DB identifier  
lab-009

Role  
Instance

CPU  
 1.83%

Current activity  
 0 Connections

Info  
 Available  
Engine  
MySQL Co

## Connectivity &amp; security

## Monitoring

## Logs &amp; events

## Configuration

## Maintenance &amp; backups

## Tags

## Connectivity &amp; security

## Endpoint &amp; port

Endpoint  
lab-009.c0gdjtl4xoqm.us-east-1.rds.amazonaws.com

Port  
3306

## Networking

Availability zone  
us-east-1a

VPC  
vpc-f2b8e188

Subnet group

## Test and Validation

Open a browser and using the EC2 instance's public IP address, connect to the database via phpMyAdmin.

http://<public IP EC2 instance>/phpMyAdmin



Welcome to phpMyAdmin

Language

English



Log in 

Username:

admin

Password:

.....

Go